Modern ML Research Sampler

A 12-week hands-on syllabus for exploring post-training, RL, agents, evals, interpretability, and ML systems

Primary goal: by the end of 12 weeks, identify the frontier area you most want to pursue for the next year - based on experiments you actually ran, not just papers you read.

Phase	Weeks	Focus
I	1-2	SFT, LoRA/QLoRA, preference learning
II	3-5	RL foundations, GRPO/RLOO, RLVR research
III	6-8	Tool-using agents, evals, agent RL
IV	9-11	Interpretability, systems, synthetic data
V	12	Mini research project

Working rules

1. Every week ends with a plot, table, or benchmark result.
2. Keep one model and one domain for most of the course so results connect across weeks.
3. Use papers and courses as references; experiments are the actual curriculum.
4. For each paper, record: main claim, strongest evidence, most important experiment, and one unanswered question.
5. Prefer many small, controlled experiments over a few large expensive runs.

Week 1: SFT and PEFT

Goal: Understand supervised fine-tuning, instruction data, LoRA/QLoRA, chat templates, token masking, and basic eval design.

Reading / study

• Read the LoRA paper

• Read the QLoRA paper

• Read LIMA

• Skim current PEFT and SFTTrainer docs

Hands-on work

• Fine-tune a 0.5B-1.5B open model with LoRA on ~1k-5k examples

• Evaluate held-out performance before vs. after SFT

• Inspect where prompt tokens are masked and where loss is applied

Deliverable: baseline table comparing base model vs. LoRA-SFT model.

Week 2: Preference Learning

Goal: Learn reward models, preference data, DPO-style objectives, KL regularization, and why preference optimization differs from ordinary SFT.

Reading / study

• Read a concise DPO introduction/paper

• Review Bradley-Terry style preference modeling

• Skim alternatives such as IPO/ORPO

Hands-on work

• Train or reproduce a small DPO-style run

• Compare SFT vs. DPO vs. SFT+DPO with everything else fixed

Deliverable: controlled comparison and one paragraph explaining where preference learning helped or hurt.

Week 3: RL Foundations for LLMs

Goal: Build the minimum RL foundation needed for modern post-training: policy, trajectory, return, advantage, policy gradient, entropy, KL, and credit assignment.

Reading / study

• Derive REINFORCE

• Review policy-gradient intuition

• Study importance sampling and KL constraints

Hands-on work

• Implement REINFORCE on a tiny environment

• Plot reward over training

• Change one hyperparameter and explain the effect

Deliverable: small RL implementation you understand end-to-end.

Week 4: RLHF, PPO, GRPO, and RLOO

Goal: Connect classic policy-gradient methods to modern LLM post-training and understand why GRPO/RLOO-style methods are attractive.

Reading / study

• Read a modern RLHF overview

• Study PPO at a conceptual level

• Read GRPO/RLOO documentation or paper summaries

Hands-on work

• Run GRPO or RLOO on a small verifiable task

• Track reward, KL, output length, and task accuracy

Deliverable: first successful RL fine-tuning run.

Week 5: RLVR as Research

Goal: Move from reproduction to hypothesis testing using verifiable rewards.

Reading / study

• Read one recent RLVR paper carefully

• Focus on reward design, ablations, and generalization

Hands-on work

• Compare sparse vs. dense vs. curriculum reward

• Measure sample efficiency, pass@1, pass@k, OOD performance, and reward hacking

Deliverable: ~2-page research note with hypothesis, experiment, results, and interpretation.

Week 6: Tool-Using Agents

Goal: Learn the agent loop: state -> action -> tool -> observation -> next action.

Reading / study

• Review a current agents course/tutorial selectively

• Study function calling and tool feedback

Hands-on work

• Build a small agent with calculator, code execution, and search/index tools

• Log complete trajectories

Deliverable: working agent plus trajectory traces.

Week 7: Agent Evaluation

Goal: Learn trajectory evaluation, deterministic graders, tool-use accuracy, failure taxonomies, and behavioral coverage.

Reading / study

• Read one modern agent-evals guide

• Review benchmark design and LLM-judge caveats

Hands-on work

• Create a small benchmark, ideally in a domain you know well

• Compare LLM alone vs. LLM+search vs. LLM+compiler/simulator or equivalent tools

Deliverable: benchmark table and failure taxonomy.

Week 8: Agent RL

Goal: Combine RL with multi-step tool-using environments; confront credit assignment and long-horizon behavior.

Reading / study

• Read one recent agentic-RL paper

• Focus on rollout structure, environment state, and reward assignment

Hands-on work

• Train on a tiny multi-step environment

• Compare outcome-only reward vs. intermediate reward

Deliverable: trajectory-level learning curves and qualitative failure analysis.

Week 9: Interpretability

Goal: Sample mechanistic/model-behavior research using a model you have already trained.

Reading / study

• Read one practical interpretability paper/tutorial

• Review probes, activation steering, and causal interventions

Hands-on work

• Probe or steer one behavior changed by SFT/RL

• Compare before vs. after training

Deliverable: one figure showing an internal or behavioral difference tied to training.

Week 10: Inference and ML Systems

Goal: Profile and optimize the same model; determine whether systems work is more engaging than training research.

Reading / study

• Review quantization, KV cache, batching, speculative decoding, and attention optimizations

Hands-on work

• Measure prefill latency, decode latency, tokens/sec, VRAM, and batch scaling

• Apply one optimization and re-measure

Deliverable: performance table before vs. after optimization.

Week 11: Synthetic Data and Distillation

Goal: Explore whether better data selection can beat simply generating more data.

Reading / study

• Read one modern synthetic-data or distillation paper

• Study filtering and diversity selection

Hands-on work

• Compare random synthetic data vs. difficulty-filtered vs. diversity-selected vs. failure-targeted data

Deliverable: data-quality ablation with fixed model and training budget.

Week 12: Mini Research Project

Goal: Pick the area that kept pulling you back and turn an observation into a small research result.

Reading / study

• Re-read the most relevant prior papers

• Write down a falsifiable hypothesis before running experiments

Hands-on work

• Observation -> hypothesis -> experiment -> ablation -> result -> explanation

• Repeat only the experiments needed to support or reject the claim

Deliverable: short research report with question, motivation, related work, setup, results, ablations, failures, and next steps.

Recommended Weekly Cadence

Day	Suggested use
Saturday	1.5 h reading + 3 h implementation + 1 h experiments
Sunday	3 h experiments + 1 h analysis + 1-2 h writing / follow-up reading

Suggested project spine

Use one small open model (roughly 1B-3B) and one technical domain across most weeks. For example: code or SystemVerilog. Week 1 SFT it, Week 2 preference-tune it, Weeks 4-5 RL-tune it, Weeks 6-8 turn it into an agent and evaluate/train it, Week 9 inspect it, Week 10 optimize inference, and Week 11 distill or improve its data.

Why this matters: a connected sequence teaches you more than twelve unrelated demos because you develop intuition for one model's behavior and can form better hypotheses.

Research notebook template

Paper / idea: _____

Main claim: _____

Why should I believe it? _____

My reproduction: _____

Result: _____

What surprised me? _____

Next experiment: _____